



Goal

Bridge the platform from exploratory analytics into AI-assisted material recommendation workflows.

KEY INSIGHTS

Material Recommendation Table (The "Shortlist")

Its shows the material selected for deeper analysis (done by AI?)

Multi-Objective Optimization Plot (The "Trade-off" Bubble Chart)

Visualize why the "Winners" were chosen. "Perfect Material" is a Giant Bubble in the Bottom-Left corner:

- Position: You want bubbles in the Bottom-Left (Low Density + Low Band Gap).
- Size: The Bigger the bubble, the more stable the material is.

AI Feature Engineering Table (The "AI Brain" View)

This table shows "Derived Features." These aren't values we measured; they are values we calculated to help an AI model learn. AI Recommendation engine will look at these specific scores to understand why a material is good, rather than just looking at raw physics

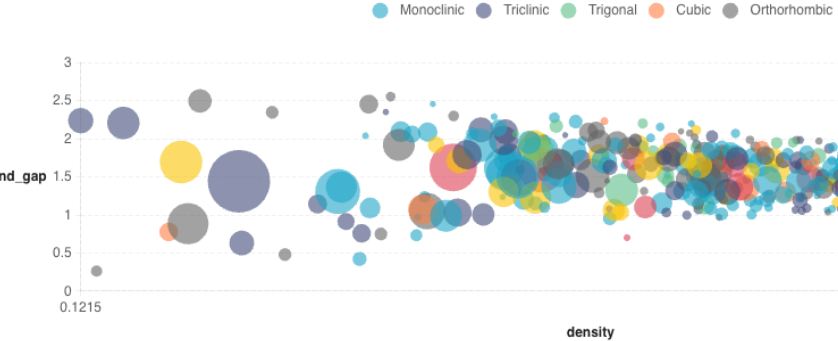
- Sustainability Score: Estimates how "green" or available the material is.
- Weight Efficiency Ratio: A custom metric for mobile applications.

Material Recommendation Table

material_id	density	band_gap	formation_energy_per_atom	is_metal
mp-1	1.801930245976053	0	0.019341450000002	1
mp-10	6.294908691696857	0	0.112397205	1
mp-100	13.41932142688018	0	0.2121797899999993	1
mp-1000	4.937715825789216	1.593	-2.003574658333334	0
mp-10000	11.182943515494184	0	-1.171399257604169	1
mp-10003	7.8163588096187855	0	-0.336685225833334	1
mp-10004	9.144741152243927	0	-0.348592201994051	1
mp-10006	7.781764325671806	0	-0.262565704583334	1
mp-10008	2.012547009593427	0	-0.044570664999999	1
mp-10009	5.154975029515741	0.7852999999999991	-0.438633351249999	0



Multi-Objective Optimization Plot



AI Feature Engineering Table

Show

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material_id ⚙	standard_formula ⚙	density ⚙	band_gap ⚙	energy_above_hull ⚙	is_metal ⚙	crystal_system ⚙	conductivity_score ⚙	stability_score ⚙	weight_efficiency_ratio
mp-12957	O2	1.974484493998944	1.0933	0	0	Monoclinic	0.9147		0.50
mp-1524462	O2	1.804134008741377	2.7061	0	0	Monoclinic	0.3695		0.55
mp-3347313	C	2.1761568056275022	1.3761999999999999	0	0	Monoclinic	0.7266		0.45
mp-568348	P	2.232614319007421	1.8454	0	0	Monoclinic	0.5419		0.44
mp-569794	Ta	:3343133	0	0		Tetragonal	N/A		0.06
mp-1949606	Cs	1.795900693546087	0	0		Tetragonal	N/A		0.55
mp-62	Pa	39614534	0	0		Tetraconal	N/A		0.06